Fiscal Year:	FY 2015	**Task Last Updated:**	FY 04/08/2015
PI Name:	Gibble, Kurt Ph.D.		
Project Title:	Contributions to the Evaluation of the ACES Clock PHARAO		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2011 Fundamental Physics NNH11ZTT001N
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No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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Task Description:	ACES (Atomic Clock Ensemble in Space) is a European Space Agency (ESA) project that will test fundamental physics, search for the time variation of fundamental constants, and enable highly precise frequency comparisons between clocks on Earth. It is planned to operate for 18 months on the International Space Station. PHARAO (Projet d'Horloge Atomique par Refroidissement d'Atomes en Orbit) is the laser-cooled, microgravity, atomic cesium clock of ACES that provides state-of-the-art accuracy, with a goal of 1 part in 10^(16) accuracy. We propose to assist in the accuracy evaluation of PHARAO, a central scientific objective of the ESA mission. We will contribute to the analysis of the three largest sources of systematic uncertainty: clock frequency shifts due to the collisions of ultracold atoms, a Doppler shift known as the distributed cavity phase shift, and a frequency shift due to the lensing of the atomic wavepackets by the clock's microwave field. The current working uncertainty budget projects 2.2×10^(-16) accuracy. The proposed work will reduce the largest contribution to the clock's uncertainty to negligible levels and investigate significant reductions of		

	the others, with a goal of reaching 10^(-16) accuracy. The work will help define the limits of PHARAO's accuracy and identify workable procedures to demonstrate its accuracy.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our work on microwave cavities produced an improved design that will be used for Cs and Rb fountain clocks. These are being incorporated into new clocks being built at the British and South Korean National Laboratories, National Physical Laboratory (NPL), and KRISS. The new cavity design allows significantly reduced clock errors due to Doppler shifts. Our work has also led to a deeper understanding of the microwave lensing frequency shift that is being used as a correction of International Atomic Time.
Task Progress:	During the last year, we have published two papers: 1) a simplified and explicit treatment of the microwave lensing frequency shift, “Ramsey spectroscopy, matter-wave interferometry, and the microwave-lensing frequency shift,” Kurt Gibble, Phys. Rev. A 90, 015601 (2014), and 2) clarified some issues related microwave lensing frequency shift and Doppler shifts (distributed cavity phase shift) in the first accuracy evaluation of the newest U.S. primary atomic frequency standard, “Comment on 'First accuracy evaluation of NIST-F2',” Kurt Gibble, Metrologia 52, 163 (2015). We completed calculations of the standing wave and loss fields with large finite element models of the PHARAO clock cavity. There is an inconsistency between the calculated resonant frequencies and the measurements. For homogeneous and sample inhomogeneous wall resistances, we have done two sets of models, one for the specified dimensions and one with dimensions altered to reproduce the measured resonance frequencies. We have calculated the effective phase for these models and the next task is to analyze the Doppler shifts, known as the distributed cavity phase errors. The phase variations and distributed cavity phase errors are expected to be similar for the two models, and therefore give confidence in our models, in spite of the disagreement of the PHARAO clock cavity dimensions and the cavity’s measured spectrum. An internal document was prepared describing our methods of calculating the PHARAO cavity fields, which will support a future publication. We have studied the microwave amplitude inhomogeneity in the PHARAO state selection cavity and find a large variation from measurements of Rabi flopping. Further measurements that change the Ramsey cavity power can accurately define the direction of the gradient in the selection cavity. The variation of the selection probability impacts the distributed cavity phase and collisional frequency shift systematic errors. For the microwave lensing frequency shift, we developed analytic and nearly analytic models for the PHARAO microwave lensing frequency shift, as well as a numerical model. We expect to soon prepare a manuscript for publication on the PHARAO microwave lensing frequency shift. We have completed a set of precision measurements of Cesium scattering phase shifts through a series of Feshbach resonances with our atomic fountain clock. We have completed a preliminary analysis of our data and constructed an end-to-end model of the experiment to establish the accuracy of the data. A number of systematic errors were studied and characterized, including cold collision frequency shifts, the dependence of the phase measurements on atomic density, off-resonant probing of unscattered atoms, the minimization of 60 Hz magnetic fields, inelastic collisions, and tests of our mathematical model. Our measurements are expected to provide precise information needed to predict the ultracold collision frequency shift of PHARAO. A number of talks were given during the past year. Two talks were given at the 2014 European Frequency and Time Forum, four talks were given at the 2015 Joint meeting of the European Frequency and Time Forum and the International Frequency Control Symposium in Denver, a talk on our PHARAO work at the NASA Fundamental Physics meeting, and a talk at Kirkland Air Force Base.
Bibliography Type:	Description: (Last Updated: 05/28/2026)
Articles in Peer-reviewed Journals	Gibble K. "Ramsey spectroscopy, matter-wave interferometry, and the microwave-lensing frequency shift." Physical Review A. 2014 Jul;90(1):015601. http://dx.doi.org/10.1103/PhysRevA.90.015601 , Jul-2014
Articles in Peer-reviewed Journals	Gibble K. "Comment on 'First accuracy evaluation of NIST-F2.'" Metrologia. 2015 Feb;52(1):163. http://dx.doi.org/10.1088/0026-1394/52/1/163 , Feb-2015